WASP-10b

R-1410

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_220108` · created 2026-08-01T05:08:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459587.6707 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.207 +/- 0.059
Transit depth (Rp/Rs)^2	4.3 +/- 2.46 [%]
Orbital Inclination (inc)	86.32 +/- 1.64
Airmass coefficient 1 (a1)	0.961 +/- 0.09
Airmass coefficient 2 (a2)	0.037 +/- 0.013
Scatter in the residuals of the lightcurve fit is	9.72 %
Best Comparison Star	#1 - [299, 182]
Optimal Aperture	3.0
Optimal Annulus	4.15
Transit Duration (day)	0.071 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.207 ± 0.059 vs 0.1592 (deviation 0.58σ)
Duration fit vs published	0.071 d vs 0.0946 d (deviation 0.62σ)
Mid-time O–C	13.9 min
Depth SNR	2.5
Residual scatter	9.72 %

Light curve

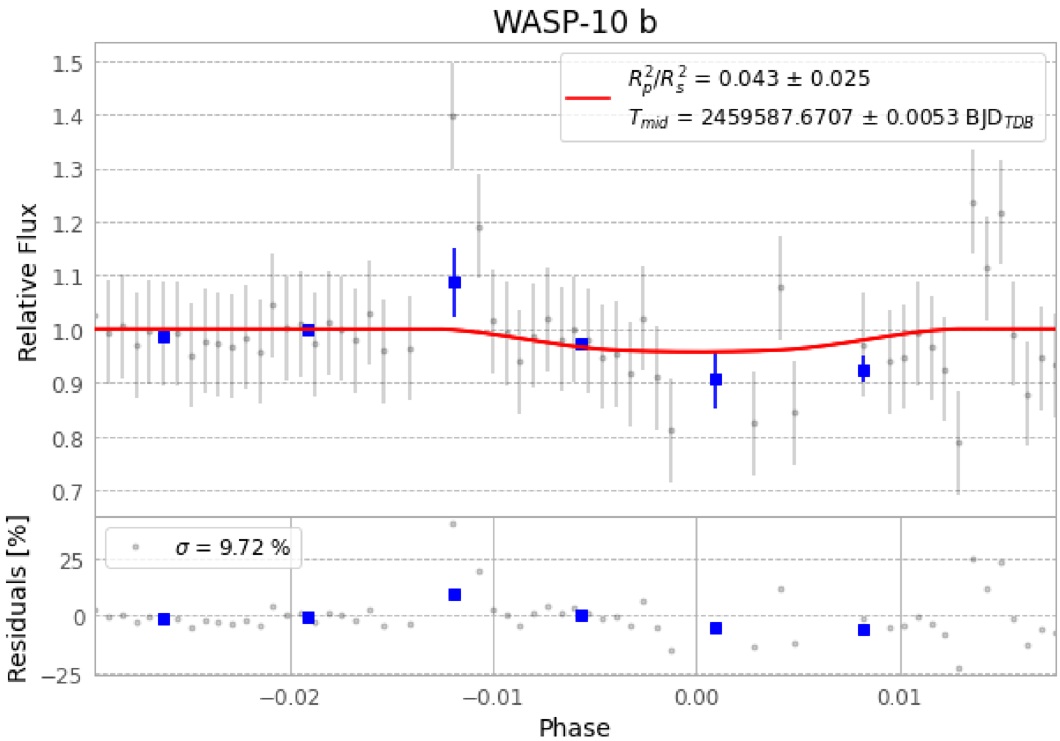

Figure 1 — FinalLightCurve_WASP-10 b_2022-01-07.png (SHA-256 496e2659f36f4689…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 282], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6a5f376172b855d4…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

83 FITS frames (55 MB), `WASP-10220108015412.FITS` → `WASP-10220108060013.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-220108.log (31a302660025…), FinalLightCurve_WASP-10 b_2022-01-07.png (496e2659f36f…), FinalLightCurve_WASP-10 b_2022-01-07.csv (552615bf2e9b…)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 05:08Z.